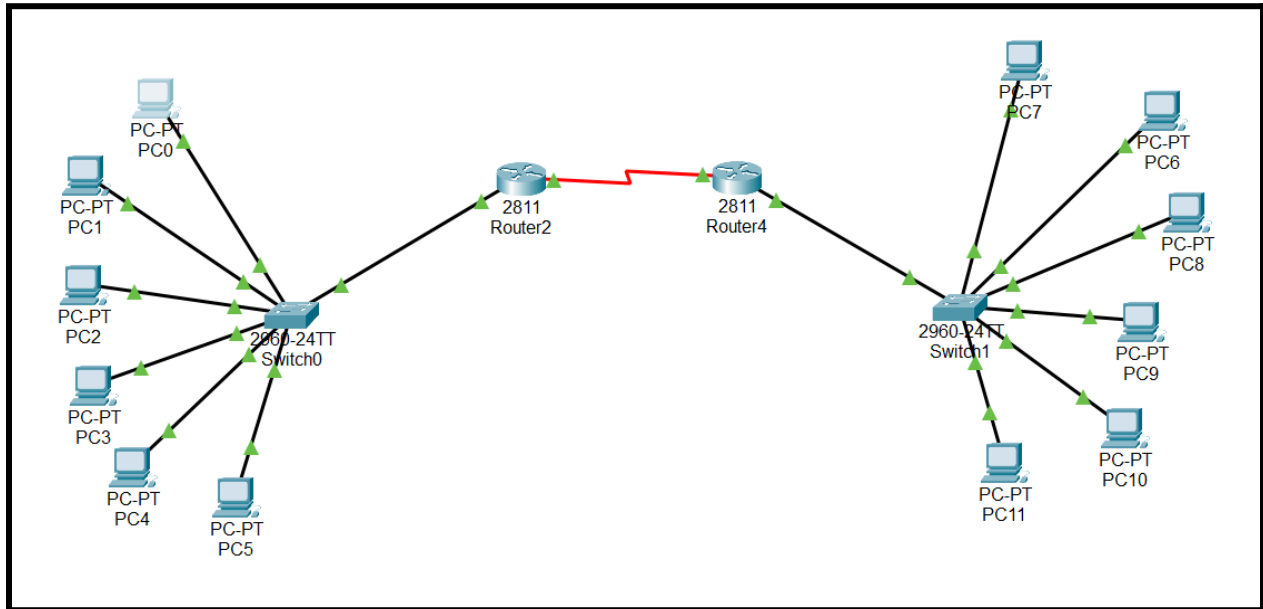


Static Routing Implementation

This report outlines the design and configuration of a network in Cisco Packet Tracer, which consists of two separate LANs connected by routers through a serial link. Static routing is implemented to facilitate communication between the LANs.

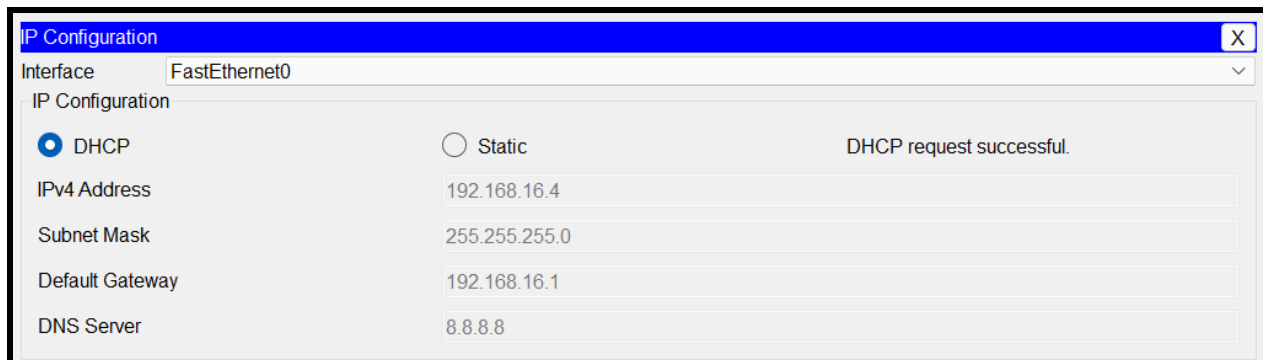


Network Overview

- LAN 1: Six PCs connected to Switch0
- LAN 2: Six PCs connected to Switch1
- Router0: Connected to LAN 1 and Router1
- Router1: Connected to LAN 2 and Router0
- Interconnection: Serial link between Router0 and Router1

Configuration

I Configured the IP through DHCP and established the connection. The connection is verified using the ping command.



On the router

IP Configuration	
IPv4 Address	172.16.1.1
Subnet Mask	255.255.0.0
Tx Ring Limit	
	10

IP Configuration	
IPv4 Address	192.168.16.1
Subnet Mask	255.255.255.0
Tx Ring Limit	
	10

```
C:\>ping 192.168.16.4

Pinging 192.168.16.4 with 32 bytes of data:

Reply from 192.168.16.4: bytes=32 time<1ms TTL=128
Reply from 192.168.16.4: bytes=32 time<1ms TTL=128
Reply from 192.168.16.4: bytes=32 time<1ms TTL=128
Reply from 192.168.16.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.16.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.11

Pinging 172.16.1.11 with 32 bytes of data:

Reply from 172.16.1.11: bytes=32 time=1ms TTL=128
Reply from 172.16.1.11: bytes=32 time<1ms TTL=128
Reply from 172.16.1.11: bytes=32 time=1ms TTL=128
Reply from 172.16.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 172.16.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Next, we need to add the serial port to the router using the following steps:

- Turn off the router
- Apply WIC-2T
- Turn on the router
- Take Serial DCE wire and connect them

Static Implementation

Open the router CLI and perform the following steps

```
%SYS-5-CONFIG_I: Configured from console by console
enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.16.0 255.255.255.0 10.0.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router(config)#ip route 172.16.1.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Verifying the connection,

We use the ping command from 192.168.16.10 to 172.16.1.10, and we are receiving packets without any loss. Hence, the connection is established between the two routers

```
C:\>ping 172.16.1.10

Pinging 172.16.1.10 with 32 bytes of data:

Reply from 172.16.1.10: bytes=32 time=15ms TTL=126
Reply from 172.16.1.10: bytes=32 time=9ms TTL=126
Reply from 172.16.1.10: bytes=32 time=20ms TTL=126
Reply from 172.16.1.10: bytes=32 time=21ms TTL=126

Ping statistics for 172.16.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 9ms, Maximum = 21ms, Average = 16ms
```